Chapter 1 Questions

1. Variables are called ----------------to the problem
- Parameters

2. Each specific assignment of values to the parameters is called an instance of the problem (True or False)
- True

3. The step-by-step procedure to solve a problem is called a(n) -------------
- Algorithm

4. A problem can be solved by only one algorithm (True or false)
- False

5. Sequential search appears to be much more efficient than binary search (T/F?)
- False

6. Algorithm analysis measures the efficiency of an algorithm as the ---------becomes large
- Input Size

7. -----------is defined as the number of times the algorithm does the basic operation for an instance of size n
- T(n)

8. W(n) is defined as the <u>minimum</u> number of times the algorithm will ever do its basic operation for an input size of n (True or False)
- False W(n)= Worst Case Complexity

9. A(n) is called average-case time complexity of the algorithm (True or False)
- True

10. Algorithms with time complexities such as n and 100n are called ----- algorithms
- Linear time

11. Algorithms with time complexities such as n^2 are called quadratic-time algorithms (True or False)
- True

12. Any quadratic-time algorithm is eventually more efficient than any linear-time algorithm (True or False)

- False

13. Function such as 5n^2 and 5n^2+100 are called ----- functions

- Pure-Quadratic

14. ---------- assumes something is true, and then does manipulations that lead to a result that is not true

- Proof By Contradiction

Chapter 2 Questions

1. Iterative algorithms execute faster than recursive ones because no …….. needs to be maintained

- Stack

2. Divide and conquer divides an instance of a problem into ………… instances

- Two or more smaller

3. Bottom-up is to get a solution to a top-level instance of a problem by going down and obtaining solutions to smaller instances (True / False)

- False (because this defines the top-down)

4. Divide-conquer-obtain are the main steps of the divide and conquer strategy (True/False)

- True

5. A sorting algorithm that does not use any extra space beyond that needed to store input is called an …………

- In-place sort

6. In quick sort all items smaller than the pivot are placed on the right of it (True/False)

- False

7. In quick sort, the partitioning of the array is done by the ……… procedure

- Partition

8. In the good condition performance of quick sort we will get O(nlogn) True/False

- True

9. Average case time complexity of quick sort is θ(nlogn) True/False

- True

10. The worst case of quick sort is $O(n^2)$ [True/False]

- True

11. Strassen's method works best only for 2X2 matrices (True or False)

- False

12. In Strassen's method, large matrices are usually divided into ……. sub-matrices

- Four

13. Strassen's algorithm is always more efficient than standard multiplication algorithm in terms of additions/subtractions (true/false)

- False Why? Only for large values of n

14. Strassen's algorithm is always more efficient than standard multiplication algorithm in terms of multiplications (true/false)

- True

15. In the multiplication of large integers, the division process is continued until ……value is reached

- Threshold

16. Two-way merging means combining two sorted arrays into one sorted array (true/false)

- True

17. ……… a sorting technique that sequences a list by continually dividing the list and then moving the lower items to one side and the higher items to the other side

- Quick Sort

18. ………….. divides the array into two halves, which are re-sorted recursively and then merged to create a sorted whole

- Merge Sort

19. …………. is a defined data type to mean an array big enough to represent the integers in the application of interest

- Large integer